

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN
HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE ORGANISATION OF
HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

National Exam of Higher National Diploma 2022 Session

Specialty/option: Software Engineering (SWE)

Paper : Computer Technology

Duration : 4 hours

Credits : 6

Instructions: Answer all questions

Section A: Programming

40marks

1. C programs are converted into machine language with the help of

- a) An Editor
- b) A compiler
- c) An operating system
- d) None of the above

2. A C variable cannot start with

- a) An alphabet
- b) A number
- c) A special symbol other than underscore
- d) both (b) and (c)

3. Which of the following is allowed in a C Arithmetic instruction

- a) []
- b) {}
- c) ()
- d) None of the above

4. HTML documents contain one root tag called

- a) HEAD
- b) Title
- c) Body
- d) HTML

5. Basic fundamental block is called
- a) html tag
 - b) html body
 - c) html attribute
 - d) html element
6. A field or column that uniquely identifies a row in a database table is a(n)?
- a) Variable
 - b) Identifies
 - c) Principal
 - d) Key
7. Which of the following is not a feature of most database management software?
- a) Sorting records in a table
 - b) Creating records
 - c) Preventing poorly designed tables
 - d) relating tables
8. before you can enter any data into a database table, you must do all of the following except?
- a) Determine the attribute the table will hold
 - b) Provide names for each attribute
 - c) Provides data types for each attribute
 - d) Determine minimum and maximum values for each attribute
9. You can extract subsets of data from database table using a(n)?
- a) Query
 - b) Sort
 - c) Investigation
 - d) Subroutine
10. A database table has the structure tblphoneorders (ordernum, custName, custphoneNum, itemOrderd, quantity). Which SQL statement could be used to extract all attributes for orders for item AB3333?
- a) `SELECT * FROM tblphoneorderd WHERE itemorderd = "AB3333"`
 - b) `SELECT tblphoneOrders WHERE itemorderd = "AB3333"`
 - c) `SELECT itemOrderd FROM tblphoneOrders WHERE = "AB333"`
 - d) Two of these are correct

11. A network administrator is building a network for a small business that has 22 hosts.

The ISP has assigned only one Internet routable IP address. Which IP address block can the network administrator use to address the network?

- a) 10.11.12.16 /28
- b) B. 172.31.255.128 /27
- c) 192.168.1.0 /28
- d) 209.165.202.128 /27

12. What subnet mask would be used with the hosts in the 128.107.176.0 /22 network?

- a) 255.0.0.0
- b) 255.248.0.0
- c) 255.255.252.0
- d) 255.255.255.0
- e) 255.255.255.252

13. You have been assigned the address block 10.255.255.224 /28 to create the network addresses for point-to-point WAN links. How many of these WANs can you support with this address block?

- a) 1
- b) 4
- c) 7
- d) 14

14. Which OSI layer is associated with IP addressing?

- a) 1
- b) 2
- c) 3
- d) 4

15. . Which type of addressing is found at the OSI Layer 2? (Choose two.)

- a) Logical
- b) Physical
- c) MAC
- d) IP
- e) Port

16. A directed graph is if there is a path from each vertex to every other vertex in the digraph.
- a) Weakly connected
 - b) Strongly Connected
 - c) Tightly Connected
 - d) Linearly Connected
17. In the traversal we process all of a vertex's descendants before we move to an adjacent vertex.
- a) Depth First
 - b) Breadth First
 - c) With First
 - d) Depth Limited
18. State True or False. i) Network is a graph that has weights or costs associated with it. ii) An undirected graph which contains no cycles is called a forest. 7/17/2017 100 TOP iii) A graph is said to be complete if there is no edge between every pair of vertices.
- a) True, False, True
 - b) True, True, False
 - c) True, True, True
 - d) False, True, True
19. Inserting an item into the stack when stack is not full is called Operation and deletion of item from the stack, when stack is not empty is calledoperation.
- a) push, pop
 - b) pop, push
 - c) insert, delete
 - d) delete, insert
20. Is a pile in which items are added at one end and removed from the other.
- a) Stack
 - b) Queue
 - c) List
 - d) None of the above

21. scheduling is approximated by predicting the next CPU burst with an exponential average of the measured lengths of previous CPU bursts.

- a) multilevel queue
- b) RR
- c) FCFS
- d) SJF

22. A process control block should contain _____

- a) the process ID
- b) locations to store register values
- c) a list of all open files
- d) All of these responses are correct

23. The _____ multithreading model multiplexes many user-level threads to a smaller or equal number of kernel threads.

- a) many-to-one model
- b) one-to-one model
- c) many-to-many model
- d) many-to-some model

24. is the number of processes that are completed per time unit.

- a) CPU utilization
- b) Response time
- c) Turnaround time
- d) Throughput

25. Which of the following devices forwards data packets to all connected ports?

- a) Router
- b) Switch
- c) Bridge
- d) Hub

26. .Of the following routing methods, which is likely to take the most amount of administration time in the long term?

- a) Static
- b) Link state
- c) Distance vector
- d) Dynamic

27. Your manager asks you to look into some upgrades for your network. The current network is a 10Base2 system, and you have been experiencing numerous hard-to-track-down cable problems. As a result, you have decided to upgrade to a 10BaseT system. On the networking vendor's price list are both active and passive hubs. The passive hubs are considerably cheaper than the active ones, and you are tempted to opt for them so that you come in under budget. A colleague advises you against the purchase of passive hubs.
28. What is the primary difference between an active and a passive hub?
- a) Passive hubs do not offer any management capabilities.
 - b) Passive hubs cannot be used in full-duplex mode.
 - c) Passive hubs do not regenerate the data signal.
 - d) Passive hubs forward data to all ports on the hub, not just the one for which they are intended.
29. Which of the following statements best describes a gateway?
- a) It is a device that enables data to be routed from one network to another.
 - b) It is a term used to refer to any device that resides at the entrance of a network.
 - c) It is a device, system, or application that translates data from one format to another.
 - d) It is a network device that can forward or block data based on the MAC address embedded within the packet.
30. Which operator is evaluated first?
- a) NOT
 - b) AND
 - c) XOR
 - d) OR
31. How many return statements are allowed in a Function Procedure?
- a) 0
 - b) 1
 - c) 2
 - d) There is no limit
32. What does your class can hold?
- a) data
 - b) functions
 - c) both a & b
 - d) none of the mentioned

33. What does inheritance allow you to do?

- a) create a class
- b) create a hierarchy of classes
- c) access methods
- d) None of the mentioned

34. Which of the following Information systems are aimed at improving the routine business activities on which all organizations depend?

- a) Management Information systems
- b) Decision support systems
- c) Transaction processing systems
- d) Management support systems

35. Which of the following strategies are adopted if information requirements are not well-defined?

- a) Rapid application development method
- b) Structured analysis development method
- c) Systems development life cycle method
- d) Prototyping method

36. Structured Programming involves:

- a) functional modularization
- b) localization of errors
- c) decentralized programming
- d) stress on analysis

37. Which of the following is not a fact-finding technique?

- a) Third party enquiry
- b) Interview
- c) Questionnaire
- d) Record reviews

38. In an E-R diagram attributes are represented by

- a) rectangle
- b) square
- c) ellipse
- d) triangle

39. In case of entity integrity, the primary key may be

- a) not Null
- b) Null
- c) both Null & not Null.
- d) any value.

40. Count function in SQL returns the number of

- a) values.
- b) distinct values.
- c) groups.
- d) columns.

SECTION B: (60MARKS)

A. Programming (20 Marks)

1. Distinguish between the following as seen in computer programming with examples if necessary
 - i. High level and low level programming language (1mk)
 - ii. C programming language and C++ (1mk)
 - iii. Program interpretation and program compilation (1mk)
2. Write a C code that computes the factorial of an integer n . When does the code fail (upper limit on n for the result to be correct) and why? (5mks)
3. Consider 2 three-dimensional vectors X and Y whose components are specified by the user and stored in 1D arrays. Write the C code that verify if the two vectors are orthogonal.
(two vectors a and b are orthogonal if their dot product $a.b$ is zero. If $a=x_1i+y_1j$ and $b=x_2i+y_2j$ then $a.b=x_1x_2+y_1y_2$) (6mks)
4. Described App Widgets? (6mks)

B. Database (15 Marks)

1. What are the characteristics of a relational database? (6mks)
2. List five disadvantages of file processing. (5mks)

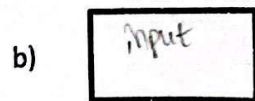
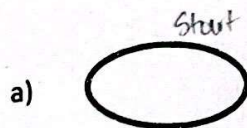
3. When would using a personal DBMS make sense? (2mks)
4. Describe what the term normalization means.
5. (2mks)

C. Networking (15 Marks)

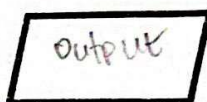
1. What are the responsibilities of Application and Presentation Layer? (4mks)
2. Give the two classes of hardware building blocks? (2mks)
3. What are the different link types used to build a computer network? (2mks)
4. Establish, if there is any, link between the OSI model layers and the TCP/IP model layers. (5mks)
5. What are the categories of Transmission media? (2mks)

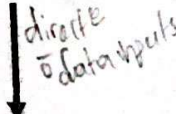
D. Algorithm and data structure 10marks

1. Define and explain the terms: Flowchart, Pseudocode, data type. (3mks)
2. Give the use of each of these symbols used in flowchart designing: (3mks)



d) 

e) 

f) 

3. Write an algorithm to find the largest value of 5 numbers in an array. (3mks)
4. Write an algorithm to search any number in an array. (3mks)